Rahul Rawat

Master's Student Data Science and Analytics | AI Agents and LLM Evaluation | Python + LangGraph + Prompt Engineering

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Enrolled through April 2027 · Available immediately · Student visa with work permit

PROFILE

Master's student in Data Science and Analytics at SRH Heidelberg, based in Mannheim, with hands on experience building, configuring, and evaluating LLM agents. I built a LangGraph multi agent RAG system with an independent LLM as Judge model scoring answers across **5 dimensions**, and a regulated credit scoring system with a plain language LLM explanation. Comfortable in Python, prompt and workflow customisation, and judging which AI use cases deliver measurable value.

SKILLS

AI and Agents	LangGraph, RAG, LLM as Judge, Prompt Engineering, Ollama, spaCy, BM25
Data and ML	Python, SQL, PySpark, scikit learn, CatBoost, XGBoost, LightGBM, Prophet, MICE
Cloud and Orchestration	GCP, BigQuery, Cloud Run, AWS, Docker, Apache Airflow, dbt, Git
Dashboards	Tableau, Looker Studio, Streamlit, Power BI
Web	React, module federation, Playwright, HTML5, CSS3

PROFESSIONAL EXPERIENCE

eRay GmbH

Data Scientist · Heidelberg · Oct 2025 to Mar 2026

- During a **6 month** eRay GmbH and SRH Heidelberg collaboration to forecast lake water quality across **4 target** indicators chlorophyll a, turbidity, pH and dissolved oxygen, built an end to end recursive time series pipeline over a **40 feature** space with a per target lag suite lag_1h, lag_24h, lag_3d, lag_7d, lag_roll_mean_24h and lag_roll_std_24h.
- Benchmarked **6 candidates** Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost and Prophet with strict tree constraints max_depth 4 and learning_rate **0.05**, landed on CatBoost MultiQuantile at alpha **0.05**, **0.5** and **0.85**, producing asymmetric **80 percent** prediction intervals that hug the 0 floor and chop the top **15 percent** of summer ghost spikes.
- Made the September evaluation defensible with a **3 pass** outlier system pH tightened from **0 to 14** down to **7.0 to 9.0**, Oct and Nov caps of **15.0** on chlorophyll a and **50.0** on turbidity, and a rolling z score at z>**2.5** over **48 hours**, and excluded 5 sparse sensors plus 3 concurrent proxies phycocyanin_abs, phycocyanin_abs_comp and toc, surfacing the honest R squared of **0.86** on dissolved oxygen and **0.81** on pH.

Technologies: *Python, CatBoost, Prophet, scikit learn, MICE*

SS Engineers and Contractors

Junior Associate Software Developer · India · Aug 2023 to Aug 2024

- With SS Engineers and Contractors running internal Data Dashboards, Analytics platforms, and Employee Portals used across the company, tasked with building and maintaining the front end features that day to day teams depended on, contributed React UI components across all three internal products, which stayed in daily use by internal teams across the company throughout the year in role.

Technologies: *React, module federation, Playwright, AngularJS, HTML5, CSS3*

SS Engineers and Contractors

Front End Developer Intern · India · Feb 2023 to July 2023

- During a six month internship at SS Engineers and Contractors, tasked with learning the codebase and then taking on small UI work across the internal Data Dashboards and Employee Portals under senior review, paired closely with senior developers to walk the codebase and then shipped focused UI components including charts, filters, and profile pages, iterating each one on code review feedback until it landed.

Technologies: *React, HTML5, CSS3, Git, code review workflow*

EDUCATION

M.Sc. Data Science and Analytics

SRH University of Applied Sciences Heidelberg, GPA 1.9 · Heidelberg · Apr 2025 to Present

Bachelor of Technology in Computer Science

GL Bajaj Institute of Technology and Management, CGPA 7.3 of 10 · Greater Noida, India · 2019 to 2023

PERSONAL PROJECTS

Multi Agent RAG with LLM as Judge and Multilingual EN and DE Support

2025

- For an English only hybrid BM25 plus dense RAG policy analysis system over a **14 document** policy corpus that could not serve German speakers, tasked with adding multilingual support without duplicating the corpus, migrated embeddings and retrieval to a paraphrase multilingual MiniLM L12 v2 shared vector space so a German query retrieves English sources and is answered in German end to end.
- With every agent in the graph re running language detection and producing inconsistent output languages, tasked with a single source of truth, built a LanguageAgent that centralises seeded language detection with a confidence floor, propagates the output language directive to every downstream agent, and routes preprocessing to language matched spaCy pipelines with a blank multilingual fallback and per chunk language recording so retrieval and evaluation can slice by language.

Technologies: *Python, LangGraph multi agent, Ollama with Mistral 7B and Qwen2.5 14B, Pinecone, paraphrase multilingual MiniLM L12 v2, spaCy, BM25*

CERTIFICATIONS

- NVIDIA: Building LLM Applications With Prompt Engineering, issued November 2025.
- AWS Academy Graduate: AWS Academy Cloud Foundations, issued July 2025.
- Google Data Analytics: Foundations, Data, Data, Everywhere, issued April 2025 on Coursera.

ACHIEVEMENTS

- USAII Global AI Hackathon 2026: Finalist at Graduate Level, awarded by the United States Artificial Intelligence Institute for innovation, technical creativity, and applied AI on real world challenges.

LANGUAGES

English: Fluent, written and spoken
German: B1, in progress